Dissertation Proposal Form

PDE4445

Please fill out this form with the details for your project ideas. Refer to the appendix for clarification regarding the various terms.

Once finalized, you will also have to consider the ethical guidelines for the project. Typically, if your evaluation strategy involves human participants, you will be required to get prior ethical approval. Based on your understanding of the project and discussions with supervisor, please select the declaration that applies to you regarding ethical approval.

Student Details:

Student Name	Jayashanka Anushan
MISIS Number	M01037028

Project Details:

Proposed Title	A Low-Cost GSM-Based Companion Assistive Robot for Elderly Care and Human–Robot Interaction Research in Developing Countries
Keywords	Elderly Care, Assistive Robotics, Fall Detection, GSM Communication, Human-Robot Interaction, MediaPipe, Assisted Living, Developing Countries, Low-Cost Robotics
Problem Definition (Maximum 500 words)	Despite significant advances in assistive robotics and environmental assisted living technologies, current solutions for elder care in developing countries are largely inadequate. Commercially available assistive robots are expensive, require a stable internet connection, and assume a level of technological familiarity that most elderly users do not possess. Wearable safety devices and smart home systems similarly rely on user compliance, continuous charging, and reliable network infrastructure, which are often unrealistic in rural or resource-constrained environments. As a result, older people living alone in such contexts are underserved by existing technological interventions. There is currently a shortage of affordable, autonomous, and context-aware robotic systems specifically designed for homes with limited infrastructure, low digital literacy, and limited financial resources. This research addresses the problem of how to design and evaluate a low-cost assistive robot that can provide safe monitoring, basic health support, and family connection without relying on continuous internet access or active user interaction.
Research Question(s)	This dissertation is guided by the following research questions: How can a low-cost mobile robot be designed to provide meaningful assistive support for elderly individuals living alone in developing-world home environments?

	• To what extent can onboard computer vision deployed on low-cost hardware achieve reliable fall detection in real-world domestic settings? • Can GSM-based communication provide a practical alternative to internet-dependent smart care systems for emergency alerts and family reassurance? • What are the practical limitations and performance trade-offs of implementing multiple assistive features within a strict component cost budget? • How can multimodal HRI be effectively studied in low-cost, GSM-based robotic platforms in developing world contexts?
Background Review (Maximum 500 words)	The field of assistive robotics has seen significant advancement in high-income countries, with solutions such as PARO (therapeutic seal robot), Pepper (social humanoid), Care-O-Bot (home assistant), and ElliQ (AI companion) demonstrating benefits in elderly care including emotional support, social engagement, and task assistance [1-4]. However, these solutions share critical limitations for developing world deployment: high cost (ranging from $5,000 to $250/month subscription), dependency on Wi-Fi connectivity, and assumption of high digital literacy among users. Fall detection systems represent a critical safety feature for elderly monitoring. Current approaches include wearable accelerometers (requiring user compliance), vision-based systems (raising privacy concerns), and floor sensor networks (requiring installation infrastructure) [5-7]. Recent advances in AI-based pose estimation using tools such as MediaPipe have enabled accurate fall detection using standard RGB cameras without wearable devices [8], but these implementations typically require high-performance computing hardware and stable network infrastructure. GSM-based communication has emerged as an alternative to Wi-Fi-dependent IoT systems for health monitoring in connectivity-limited environments [9-11]. Research demonstrates that GSM SMS can deliver reliable alerts without internet access, making it suitable for rural and underdeveloped regions. However, integration of GSM with autonomous robotic systems for elderly care remains underexplored in the literature. Robot navigation in indoor environments has been well-established using SLAM (Simultaneous Localization and Mapping) algorithms in ROS (Robot Operating System) [12-14]. Studies comparing GMapping and RTAB-Map demonstrate reliable indoor localization, though computational requirements limit deployment on low-cost hardware. Human-Robot Interaction (HRI) research emphasizes the importance of perceived usefulness, ease of use, safety, and emotional connection in robot acceptance among elderly users [15-17]. Studies indicate that elderly users prefer robots requiring

	minimal interaction for core functions, supporting the zero-interaction design philosophy. The intersection of low-cost robotics, GSM communication, and developing world elderly care remains an under-researched area. This dissertation addresses this gap by integrating these elements into a single, coherent system
Aims	The primary aim of this research is to design, build, and evaluate a low-cost, GSM-based companion assistive robot (CARA) that provides safe monitoring, basic health support, and family connectivity for elderly individuals living alone in developing countries, without requiring continuous internet access or active elderly user interaction. This aim will be pursued through the development of an integrated robotic system combining AI-based computer vision, microcontroller real-time control, and cellular communication specifically optimized for low-resource deployment contexts.
Objectives (Bullet points)	• To design and implement an autonomous indoor navigation system using onboard camera-based and ultrasonic sensors for obstacle avoidance, achieving collision-free navigation in at least 80% of trials in a single-floor indoor environment. • To develop and validate an AI fall detection system using MediaPipe pose estimation on NVIDIA Jetson Nano, achieving minimum 80% detection accuracy with a false positive rate below 2 per hour across 20 standardized trials. • To implement a GSM-based emergency alerting system delivering SMS notifications within 30 seconds of fall detection, operating without Wi-Fi dependency, validated across 15 test trials. • To design and build a health management subsystem including scheduled medication reminders, RFID-based confirmation, and automated family SMS alerts for missed doses. • To evaluate the complete CARA system against defined user requirements using structured testing protocols and critically analyze performance against each stated objective.
Brief Methodology (Include Gantt chart)	The research will follow a design science research approach consisting of six phases: Phase 1: Requirements Analysis (Weeks 1) System requirements will be defined based on literature review and user needs analysis. Functional and non-functional requirements will be documented. Phase 2: System Design (Weeks 2) Hardware architecture (Jetson Nano, Arduino Mega, sensors, actuators) and software architecture (ROS nodes, state machine) will be designed.

	Phase 3: Implementation (Weeks 3-8) Hardware assembly, firmware development, AI integration, and GSM setup will be completed incrementally. Phase 4: Individual Feature Testing (Weeks 9) Each feature will be tested independently using standardized trial protocols. Phase 5: System Integration Testing (Weeks 10-11) All features will be integrated and tested as a complete system. Phase 6: Evaluation and Documentation (Weeks 12) Final evaluation against objectives, analysis, and dissertation writing.
Evaluation Strategy	This research will employ a mixed-methods evaluation approach combining quantitative performance metrics with qualitative observations. **Quantitative Evaluation:** • Fall detection accuracy: Percentage of correct fall identifications across 20 standardized trials • False positive rate: Number of false alerts per hour during continuous monitoring • SMS delivery time: Time between fall detection and SMS received (target: <30 seconds) • Navigation success rate: Percentage of collision-free navigation trials (target: ≥80%) • Medication reminder success: Percentage of successful reminders and confirmations **Qualitative Evaluation:** • System stability observations during continuous operation • User interaction feedback (greetings, LCD messages) • Ease of remote monitoring access via browser **Testing Protocol:** • Individual feature testing under controlled laboratory conditions • Single-user standardized trials (researcher as test subject) • Obstacle avoidance stress testing with varying configurations
Ethical Declaration (i) or (ii) (Refer below)	(i) I have established that my study does not require additional human participation. - The evaluation strategy uses a single adult researcher (the author) as the test subject following standardized trial protocols. No additional human participants are involved. The research does not involve vulnerable populations, personal data collection, or observation of third parties. (ii) I agree to re-apply for approval if the nature or goals of my project change.

Deliverables (Bullet points)	• **Functional CARA Robot**: Fully assembled mobile robot with all 15 features operational • **Technical Documentation**: Complete circuit diagrams, firmware code, and software architecture • **Fall Detection System**: MediaPipe-based AI fall detection running on Jetson Nano • **GSM Alerting System**: Wi-Fi-independent SMS emergency alerting functional • **Health Management System**: Medication reminder and RFID confirmation system • **Testing Results:** Structured evaluation data across all features • **Dissertation Report:** Complete MSc thesis documenting research process and findings
Resources Needed	**Hardware:** – NVIDIA Jetson Nano 4GB (AI processing) – Arduino Mega 2560 (microcontroller) – 720p USB Webcam (camera) – HC-SR04 Ultrasonic Sensors × 6 (obstacle detection) – HC-SR501 PIR Sensor (person detection) – SIM800L GSM Module (SMS communication) – DC Gear Motors 6V × 150RPM × 4 (mobility) – L298N Motor Driver (motor control) – MFRC522 RFID Reader (medication confirmation) – DS3231 RTC Module (timekeeping) – 16×2 LCD Display with I2C (status display) – SG90 Servos × 2 (pan-tilt head) – LiPo Battery – LM2596 Buck Converters × 2 (power regulation) **Software:** – Ubuntu 18.04 (Jetson Nano OS) – MediaPipe Pose (AI fall detection) – Arduino IDE 2.x (firmware development) – Python 3.x (AI processing scripts) – Flask (web streaming) **Laboratory Facilities:** – Robotics laboratory with indoor testing space – Power supply and soldering equipment – Computing workstations Estimated Budget: USD $450-500
References	[1] T. Shibata and K. Wada, "Robot therapy: A new approach for mental healthcare of the elderly," International Journal of Social Robotics, vol. 3, no. 2, pp. 57-68, 2011. [2] K. Wada and T. Shibata, "Living with seal robots," IEEE Transactions on Robotics, vol. 23, no. 5, pp. 972-980, 2007. [3] R. Kittmann et al.,

	"Let me Introduce Myself: I am Care-O-bot 4," in Proc. IEEE International Conference on Robotics and Automation, 2015. [4] M. Tonkin et al., "Design methodology for the UX of HRI," in Proc. ACM/IEEE International Conference on HRI, 2018. [5] A. K. Bourke et al., "Evaluation of a threshold based tri-axial accelerometer fall detection algorithm," Gait & Posture, vol. 26, no. 2, pp. 194-199, 2007. [6] C. Rougier et al., "Robust video surveillance for fall detection," IEEE Transactions on Circuits and Systems for Video Technology, vol. 21, no. 5, pp. 611-622, 2011. [7] S. Liu et al., " Vision-based Fall Detection Systems," in Proc. IEEE International Conference on Image Processing, 2020. [8] S. Saraswat and G. Malathi, "Pose Estimation based Fall Detection system using MediaPipe," in Proc. IEEE International Conference, 2024. [9] M. Arun et al., "IoT-Based Patient Health and Fall Monitoring System with GSM Alerts," IJRASET, 2025. [10] G. M. Hassan, "A Real-Time Monitoring System for the Elderly Based on the GSM Network," Journal La Multiapp, vol. 3, no. 6, pp. 287-293, 2023. [11] Y. N. Al Husaini et al., "IoT Based Remote Health Monitoring System," International Journal of Biology and Biomedicine, vol. 7, no. 2, pp. 21-25, 2022. [12] T. Alhmiedat et al., "A SLAM Based Localization and Navigation System for Social Robots," Machines, vol. 11, no. 2, 2023. [13] J. Zhao et al., "Autonomous Navigation for Mobile Robots Based on SLAM," Sensors, vol. 22, no. 11, 2022. [14] B. M. F. Silva et al., "Mapping and Navigation for Indoor Robots under ROS," Preprints, 2019. [15] E. Broadbent et al., "Acceptance of healthcare robots for the older population," International Journal of Social Robotics, vol. 1, no. 4, pp. 319-330, 2009. [16] R. Beer et al., "Socially assistive robots in elderly care," Journal of the American Medical Directors Association, vol. 12, no. 2, pp. 114-120, 2012. [17] G. Demiris and H. Hensel, "Technologies for an aging society," Journal of the American Medical Informatics Association, vol. 15, no. 1, pp. 42-55, 2008.

Signature: JAYASHANKA
Date: 29/June/2026

Ethical Approval Guidelines

Declaration A

(i) I have established that my study does not require additional human participation.
(ii) I agree to re-apply for approval if the nature or goals of my project change.

Declaration B

Project goals involve human participation:

(i) My study involves human participation through:
- Observation
- Questioning

(ii) Participants will be selected without coercion.
(iii) I will obtain written informed consent from each participant.
(iv) I have arrangements in place for the protection of personal data.
(v) I agree to re-apply for approval if the nature or goals of my project change.

To be completed by the supervisor:

Supervisor Name:

Is the proposal acceptable? (YES/NO):

If not, please provide feedback:

Signature:
Date:

Appendix: Explanation of Typical Terms Used in Research Proposals

- **Title:** Write a title which briefly describes the research problem and your approach to it.

- **Keywords:** To give a clear and concise description of the scope and nature of the report, such as the main variables to be considered.

- **Problem definition:** Correctly defining the problem is the crucial first step in the research process. If the research problem is defined incorrectly, the research objectives will also be wrong. Problems must be stated in terms of underlying causes – they must be structured in a way in which they can lead to a solution. It is critical that the statement be useful for development and evaluation of potential solutions.

- **Research Question:** Indicate what you want to know most and first out of your research. These are typically written in a 'question' format, for example, 'Does a robotic exoskeleton provide an efficient solution to physical rehabilitation of the upper arm for stroke patients?' Multiple research questions should be put in a list. Questions should not be answered by a simple YES or NO, they must generate a discussion.

- **Background Review:** Explain the technical/discipline area you will be working in, the problem area that you will be addressing in your research, and where you would locate your intended work in relation to previous researchers or existing solutions.

- **Aims:** They represent the changes you hope to achieve as a result of your work.

- **Objectives:** These are the activities you undertake and the methods you propose to bring these changes about. They should be in a bulleted list, and more specific than the aims. In fact, this list should be a way of breaking down the aims into more achievable tasks. Anywhere between 4 – 7 objectives are usually considered appropriate for a project of this scale.

- **Methodology:** A brief description of the steps to be followed to achieve the objectives that have been set out. This section should describe the development plan.

- **Gantt chart:** A Gantt chart is a horizontal bar chart used in project management to visually represent a project plan over time.

- **Evaluation Strategy:** Methods and techniques used in evaluating the project. There are two basic approaches, quantitative and qualitative. Quantitative involves the generation of data, through experiments or simulation, in quantitative form, which can then be analysed. Qualitative is concerned with subjective assessment of opinions, behaviour, impressions via, for example, interviews.

- **Deliverables:** Defining your intended outcomes (be as specific as possible). It is part of good project planning. Deliverables are linked to your aims.

- **Resource Needed:** Information on the hardware, software, and research resources you will be using during your research. Be specific – Give the name of any software and hardware, rather than stating something generic.